

Chang Jin

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Education

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| Tongji University
<i>Master of Computer Science</i>
◦ GPA: 4.84 /5.00 | <i>Shanghai, China</i>
<i>Sep 2023 - Present</i> |
| École Polytechnique Fédérale de Lausanne (EPFL)
<i>Exchange Master Program of Computer Science</i>
◦ GPA: 5.62 /6.00 | <i>Lausanne, Switzerland</i>
<i>Feb 2025 - Jul 2025</i> |
| Tongji University
<i>Bachelor of Computer Science</i>
◦ GPA: 4.91 /5.00
◦ Rank: 3 /121 | <i>Shanghai, China</i>
<i>Sep 2018 – Jun 2023</i> |

Publications

- When Silence Is Golden: Can LLMs Learn to Abstain in Temporal QA and Beyond?**
 Xinyu Zhou, **Chang Jin** (co-first author), Carsten Eickhoff, Zhijiang Guo, Seyed Ali Bahrainian
 Submitted to *ICLR 2026*
- Effective and Explainable Molecular Property Prediction by a Chain-of-Thought Enabled LLM and Multi modal Molecular Information Fusion**
Chang Jin, Siyuan Guo, Shuigeng Zhou, Jihong Guan
 Published in *Journal of Chemical Information and Modeling* (JCR Q1)
- M3-20M: A Large-Scale Multi-Modal Molecule Dataset for AI-driven Drug Design and Discovery**
 Siyuan Guo, Lexuan Wang, **Chang Jin**, Jinxian Wang, Han Peng, Huayang Shi, Wengen Li, Jihong Guan, Shuigeng Zhou
 Published in *Journal of Bioinformatics and Computational Biology*
- Enhanced Adaptive Graph Convolutional Network for Long-Term Fine-Grained SST Prediction**
 Han Peng, **Chang Jin** (co-first author), Wengen Li, Jihong Guan
 Published in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (JCR Q1)
- On Evaluating the Predictability of Sea Surface Temperature Using Entropy**
Chang Jin, Han Peng, Hanchen Yang, Wengen Li, Jihong Guan
 Published in *Remote Sensing* (JCR Q1)

Research Experience

- Research on Chain-of-Thought Monitoring for Large Language Models** *Sep 2025 – Present*
Advisor: Dr. Xingcheng Xu, Dr. Qiaosheng Zhang (Shanghai Artificial Intelligence Laboratory)
 ◦ Conduct research on chain-of-thought (CoT) monitoring for large language models, aiming to improve the faithfulness and safety of reasoning processes.
 ◦ Explore methods to detect and mitigate obfuscated or adversarial reasoning traces, and develop monitoring frameworks for reliable model behavior under different optimization pressures.
- Research on Enhancing Abstention Mechanisms of Large Language Models for Temporal Reasoning Tasks** *Sep 2024 – Present*
Advisor: Dr. Ali Bahrainian, Prof. Carsten Eickhoff (Joint Lab, University of Tübingen & Brown University)
 ◦ Developed reinforcement learning based techniques to enhance LLM understanding of temporal knowledge.

- Improved model abstention mechanisms, enabling LLMs to abstain from answering unanswerable queries, reducing hallucinations and errors.
- **Research Output:** Co-first author: a manuscript submitted to ICLR 2026.

Research on Constrained Decoding Methods for Structured Text Generation in Large Language Models

Feb 2025 – Present

Advisor: Prof. Robert West (EPFL)

- Investigated capabilities of various LLMs to generate JSON objects that adhere to given JSON Schemas.
- Analyzed probability distributions during constrained decoding to reveal how schema constraints affect LLM generation behavior.
- **Research Output:** Contributed a JSON Schema-based evaluation task to the open-source project **Im-evaluation-harness (9.5k stars)**, enabling structured-output LLM evaluation.

Research on Drug Molecule Discovery Using Large Language Models

Sep 2023 – Sep 2025

Advisor: Prof. Jihong Guan (Tongji University) & Prof. Shuigeng Zhou (Fudan University)

- Constructed M3-20M, a large-scale multi-modal molecular dataset with over 20 million molecules.
- Developed LLM-MPP, a Chain-of-Thought enhanced multimodal LLM for molecular property prediction, achieving state-of-the-art performance in prediction accuracy and interpretability.
- **Publications:**
 - First author: “**Effective and Explainable Molecular Property Prediction by a Chain-of-Thought Enabled LLM and Multimodal Molecular Information Fusion**”, *Journal of Chemical Information and Modeling (JCR Q1)*.
 - Co-author: “**M3-20M: A Large-Scale Multi-Modal Molecule Dataset for AI-driven Drug Design and Discovery**”, *Journal of Bioinformatics and Computational Biology*.

Research on Spatio-temporal Modeling

Mar 2021 – Jun 2023

Advisor: Prof. Jihong Guan & Prof. Wengen Li (Tongji University)

- Developed EA-GCN, a spatio-temporal deep learning model for long-term fine-grained sea surface temperature (SST) prediction, achieving state-of-the-art performance.
- Proposed a temporal-correlated entropy method to evaluate SST predictability from global and local perspectives, aiding marine and climate monitoring.
- **Publications:**
 - Co-first author: “**Enhanced Adaptive Graph Convolutional Network for Long-Term Fine-Grained SST Prediction**”, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JCR Q1)*.
 - First author: “**On Evaluating the Predictability of Sea Surface Temperature Using Entropy**”, *Remote Sensing (JCR Q1)*.

Honors and Awards

Outstanding Graduate of Tongji University	Jun 2023
First-Class Scholarship of Tongji University (top 5%)	2019, 2021
Second Prize of China Undergraduate Mathematical Contest in Modeling, Shanghai	2020

Skills

Programming Languages: Python, C++, C

Python Packages: PyTorch, TensorFlow, PyG, sklearn and other packages related to deep learning.

English Skills: IELTS: 7.5 (Listening: 7.5, Reading: 8.5, Writing: 7.5, Speaking: 6.5)